



*JSC "D.V. Efremov Institute of Electrophysical Apparatus"
The B.P. Konstantinov Petersburg Nuclear Physics
Institute, the NRC "Kurchatov Institute"*



Physical Start-up of the CC-80 Isochronous Cyclotron

**Yu.N. Gavrish, A.V. Galchuck, S.V. Grigorenko, A.N. Kuzhlev, V.G.Mudrolyubov,
D.A. Amerkanov, S.A.Artamonov, E.M.Ivanov, G.F.Riabov, V.I.Yurchenko**

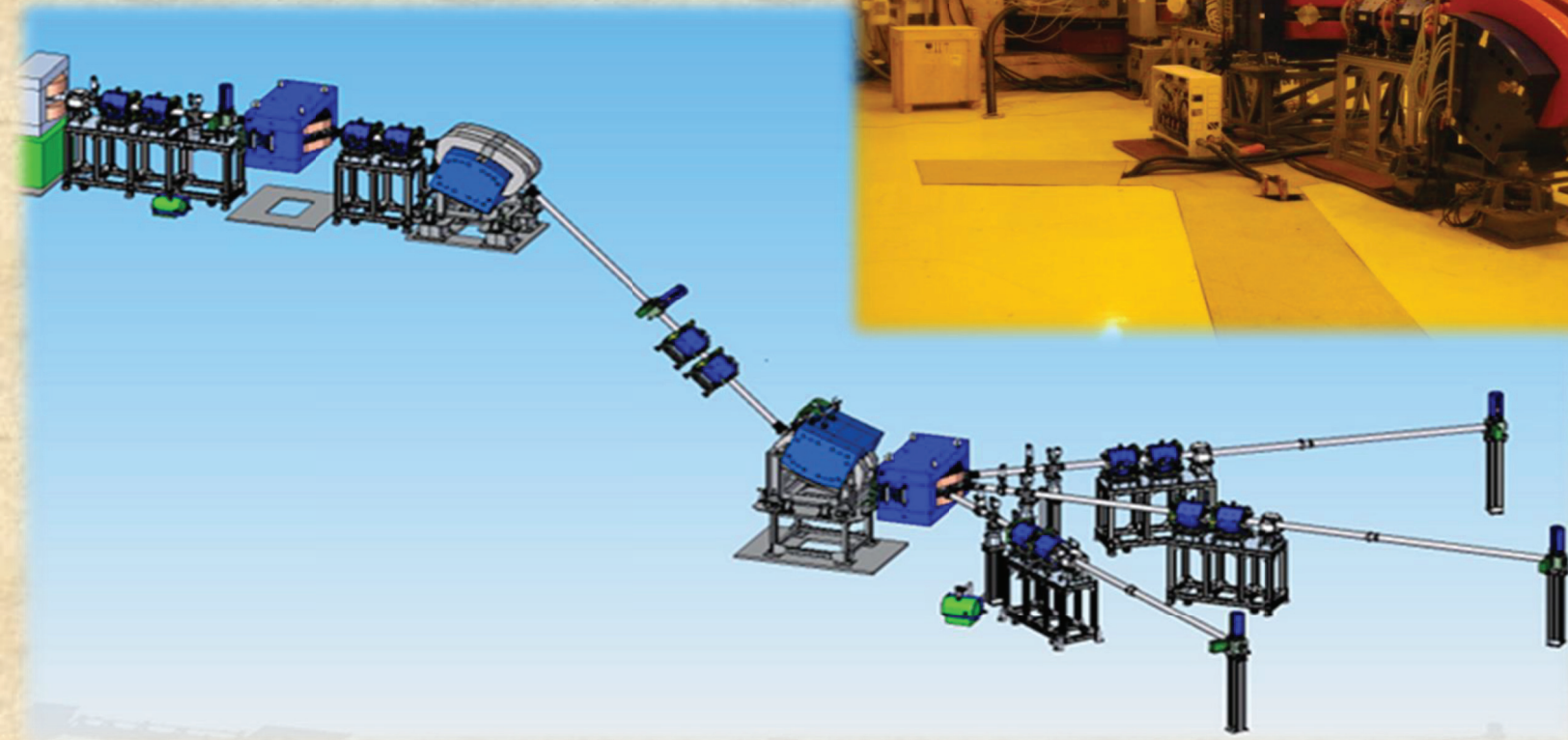
Cyclotron system CC-80

Works on the installation of a cyclotron system for the acceleration of H^- ions at energies ranging from 40 up to 80 MeV have been completed in the B.P. Konstantinov Petersburg Nuclear Physics Institute (PNPI), the National Research Centre “Kurchatov Institute”. The cyclotron is intended for production of a wide assortment of radioisotopes including radiation generators (Sr-Rb, Ge-Ga) for medicine, proton therapy of ophthalmic diseases, tests of radioelectronic components for radiation resistance and studies in the field of nuclear physics and radiation material science.

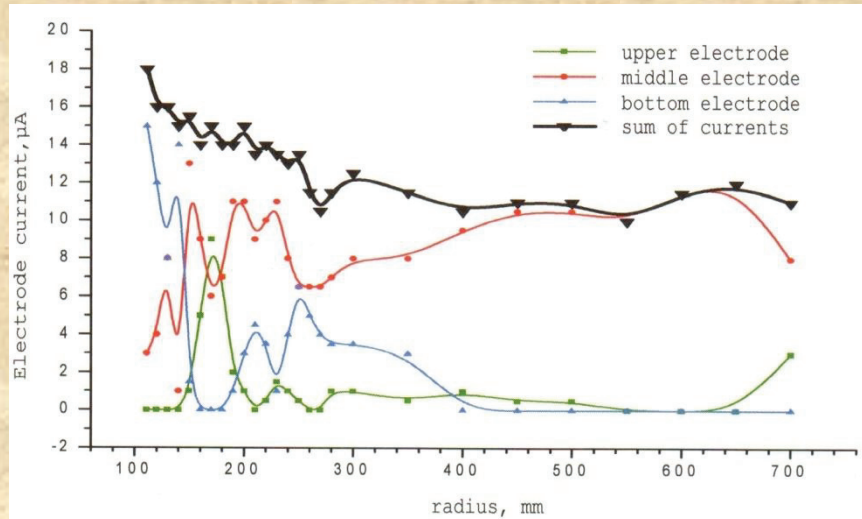
The main characteristics of the cyclotron

System, parameter	Characteristic, value
Type of accelerated particles	H^-
Type of extracted particles	H^+
Beam energy, variable, MeV	40...80
Beam current, μA .	100

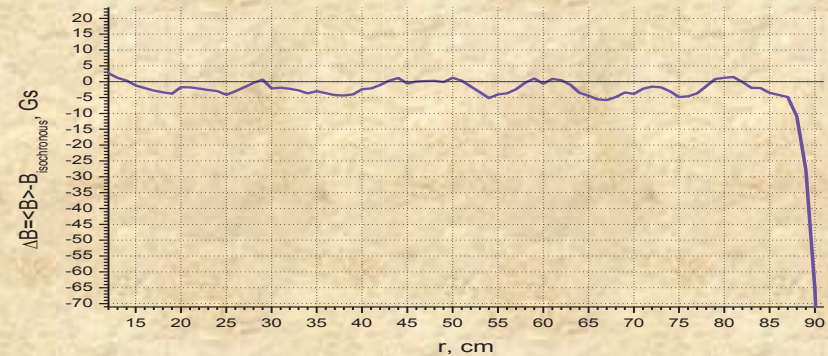
The cyclotron system equipment with the transport system of an accelerated proton beam to remote target stations is mounted in the experimental hall of building 2 and in its basement. The equipment of the cyclotron and that of the first section of the beam transport system is located on the first floor (see Fig.1); the external injection system, the RF generator and the system for the beam transport to three targets are mounted in the basement



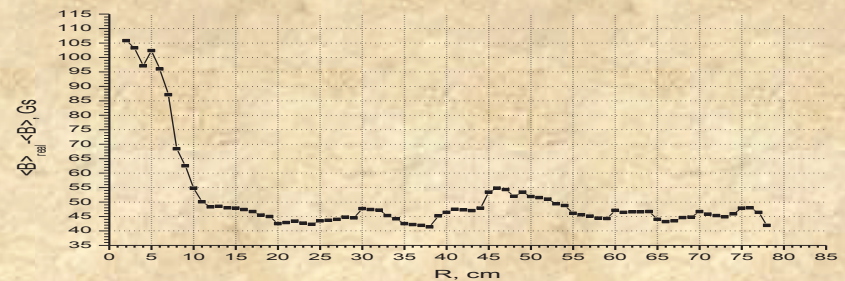
The major unit of the cyclotron, an electromagnet, was designed using a model of the magnet of the synchrocyclotron operating in PNPI and further updated. Such a decision limited significantly the choice of engineering solutions when designing the cyclotron. In the process of commissioning works some structural deficiencies made at the design and manufacturing stages were detected and eliminated.



Results of current and beam position measurements in the process of acceleration



Difference between the formed magnetic field and calculated isochronous magnetic field (prior the vacuum chamber assembly).



Difference between the actual magnetic field and formed isochronous magnetic field (after the assembly of the vacuum chamber).

The resonance acceleration system consists of two symmetrical quarter-wave resonators and is placed completely inside the vacuum chamber. The system is equipped with a capacitor for frequency tuning, AFT trimmer and RF probe. The operating frequency of RF oscillations is 41.2 MHz, which corresponds to the 2nd harmonic of the ion revolution frequency. The normal value of the RF voltage amplitude is 60 kV.



When manufacturing the vacuum chamber, a required rigidity of covers was not provided, and this resulted in smaller gap between them and deformation of resonator claddings in the process of pumping-down. This made necessary changing of vertical sizes of dee stems and resonator tanks with a subsequent radiotechnical tuning of the system. Under tests of the RF-system at a high power level, an appreciable heating of a ceramic unit of the trimmer stem took place, and its illumination and subsequent destruction was observed, which resulted from a poor choice of the trimmer position in the tank of one of resonators.

A new unit was designed, manufactured and installed, which allowed the trimmer location outside the resonator volume, and thus its heating was excluded.

The RF power supply system consists of a stabilization and control module and RF-power amplifier. The stabilization and control module designed in the Efremov Institute provided reliable operation in the process of generating 41.2 MHz operating frequency, tuning and stabilization of the natural frequency of the resonance system and accelerating voltage amplitude.

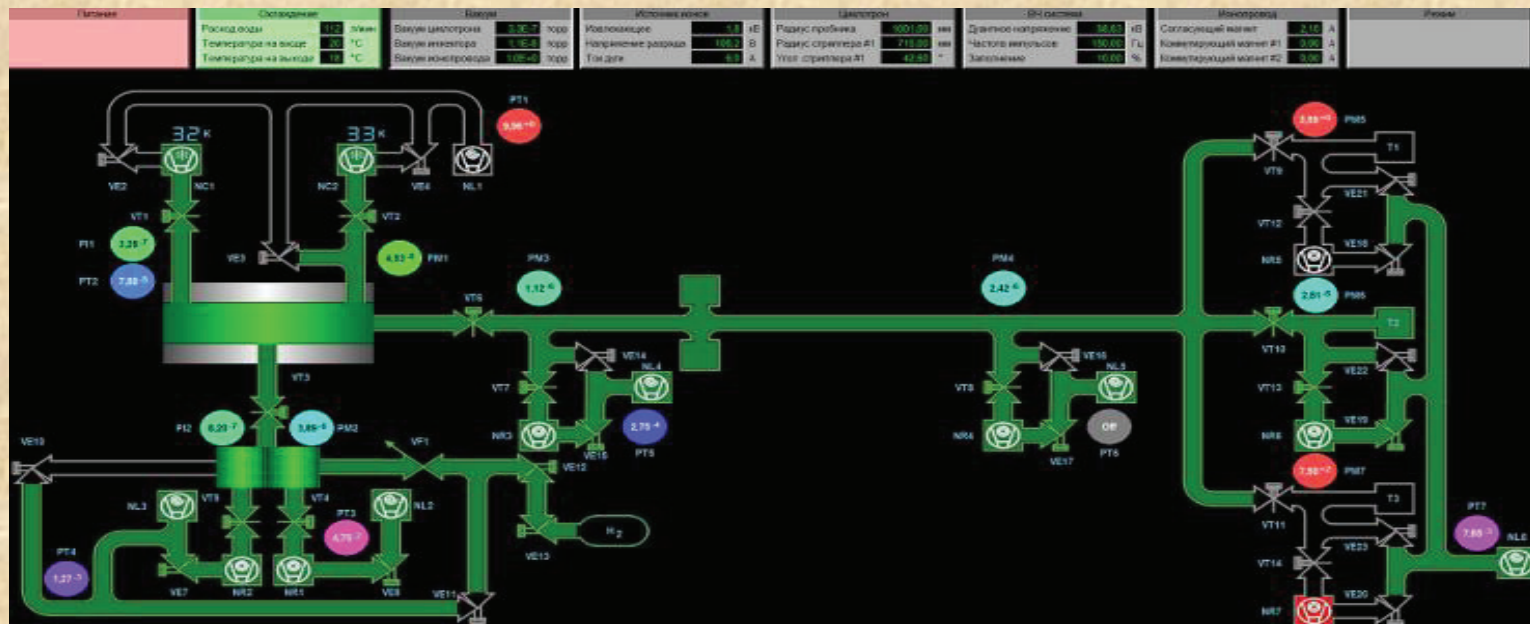


Works on updating the RF power amplifier slowed down considerably the completion of commissioning works and demonstrated a pressing need for using our own efforts to design proper amplifiers for our cyclotrons.

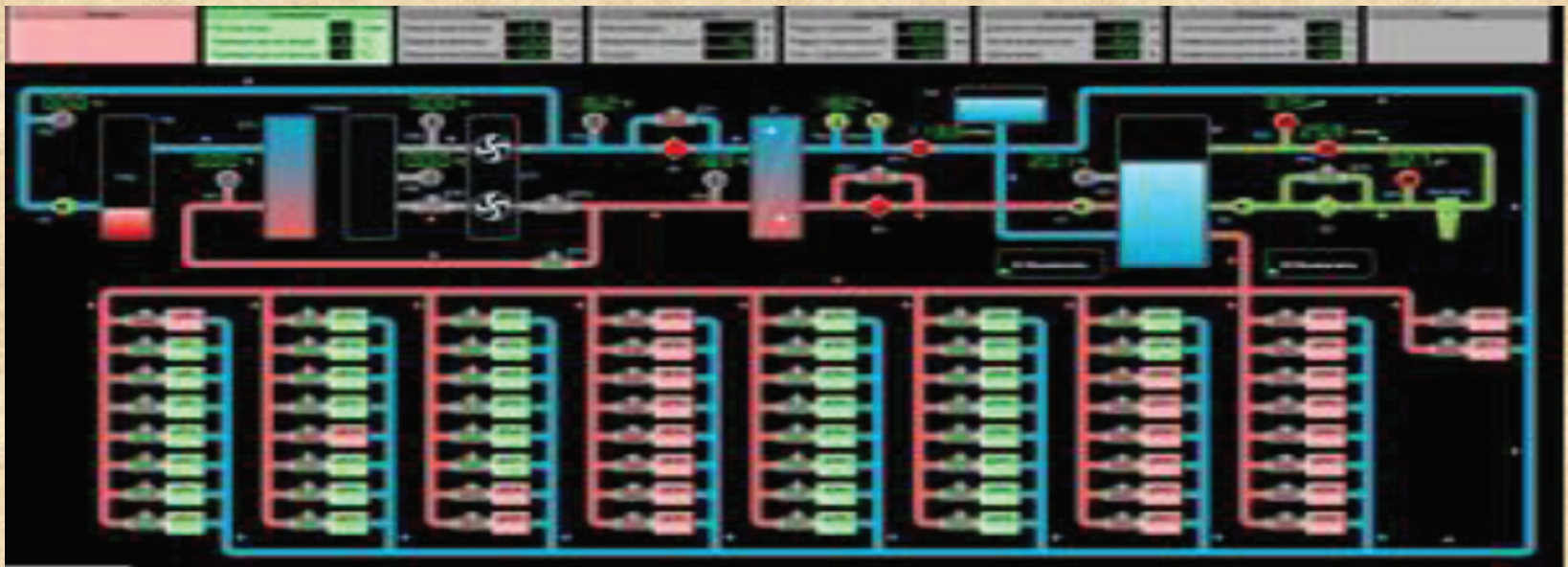
The RF power amplifier designed and delivered by the «Coaxial Power Systems», Great Britain, should provide an output power of 80 kW at a frequency of 41.2 MHz. From the results of initial tests of this equipment in operating modes, a pressing need for fundamental updating was identified. Six additional fans, each 0.25 kW, and additional partitions were installed; perforations were made in available partitions, which provided an optimal distribution of air flows and required thermal mode. The insulators used by the CPS firm in the final stage were completely replaced because of their systematic breakdowns. We detected and eliminated the whole series of errors in circuitry which did not allow a necessary stability of the output signal to be obtained and prevented normal operation of the RF system:

- the high-voltage input of the final stage was re-designed, re-manufactured and newly installed; the leading-in power cable was replaced for a shorter one;
- the circuit of transistor amplifiers' power supply was modified and one of power supply units was replaced for a more powerful;
- voltage filters of the final stage anode were replaced;
- the switch-on time of the power supply system of tube filaments was changed to improve their operation stability,
- inspection of cable inter-connections in the amplifier was conducted with subsequent replacement where necessary, etc.

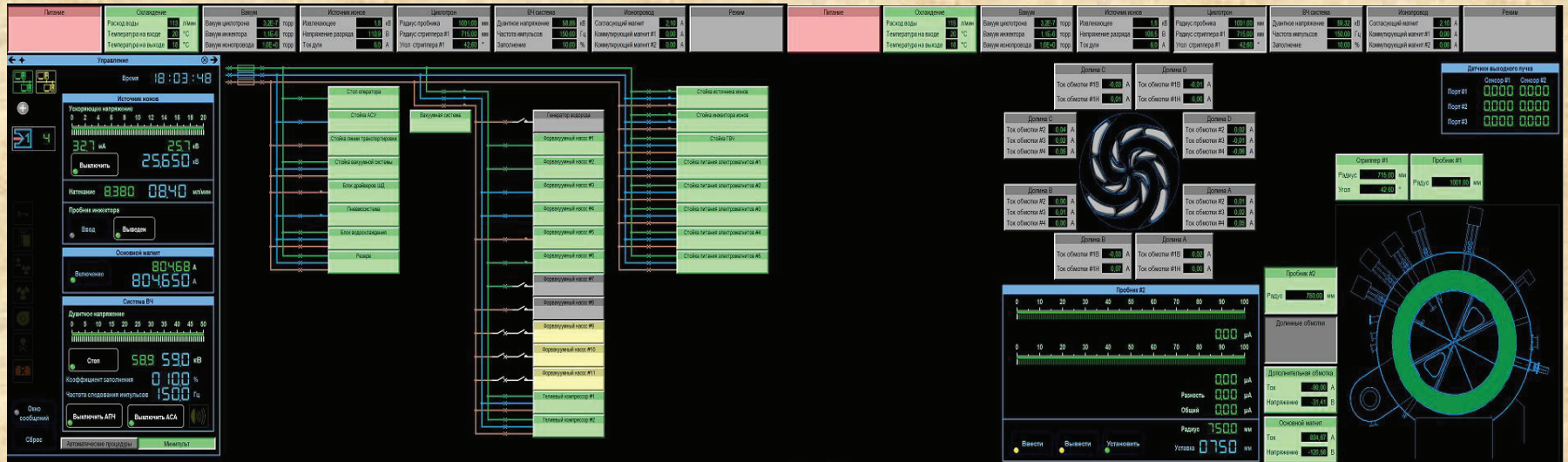
A distributed automatic control system (ACS) is used. It consists of Mitsubishi and Fastwel IO controllers and computers, each being responsible for the control of one or several sub-systems of the cyclotron. The main unit of the control system is an industrial (host) computer, which inquires slave controllers and transmits the information acquired to computers of the operator's workstation; receives commands from the operator's workstation and performs their arbitration and distribution. Data exchange is realized via network interfaces. To ensure normal operation of the automated control system, there must be a reliable special grounding to provide electromagnetic compatibility of the cyclotron equipment. At the initial stage of works, this grounding did not meet the requirements, which resulted in failures of several controllers and made us to prolong the time for carrying out commissioning works.



Status of the vacuum system units and basic parameters displayed on the operator monitor



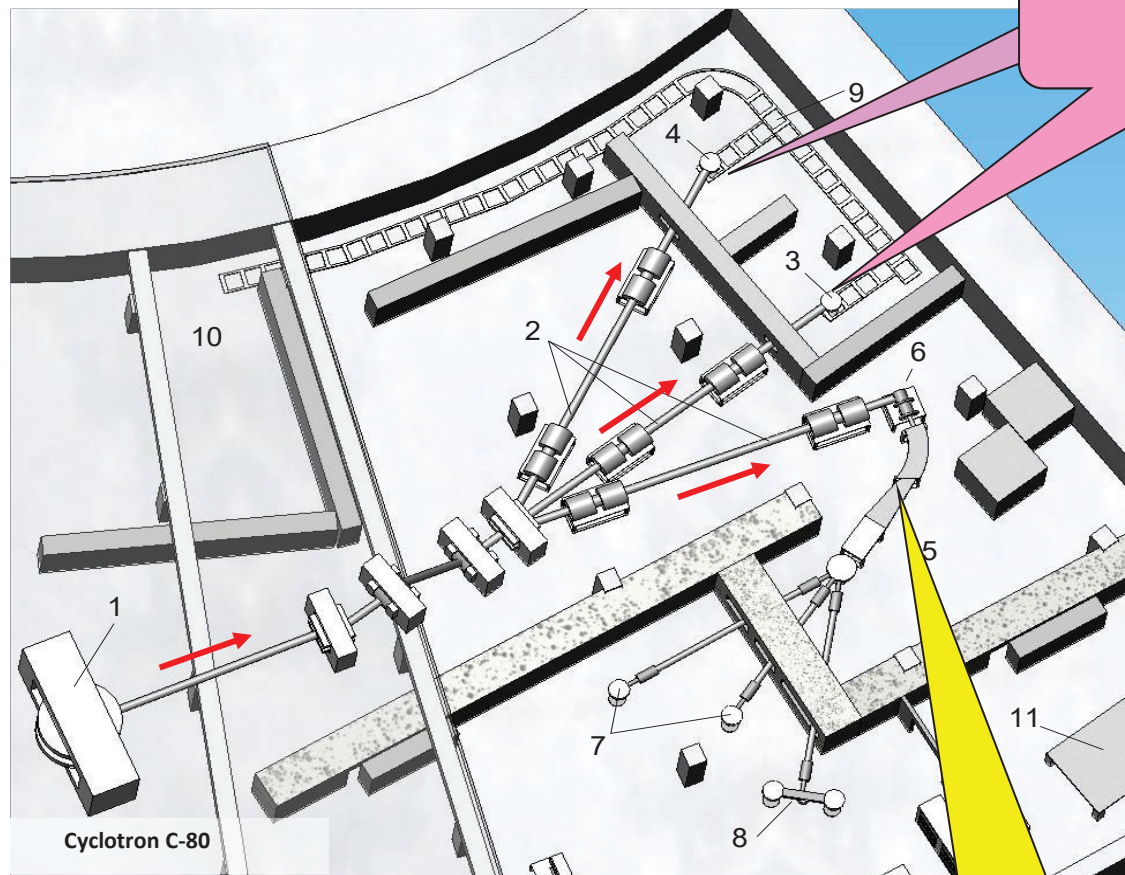
Status and basic parameters of thermo-stabilization system displayed on the operator monitor



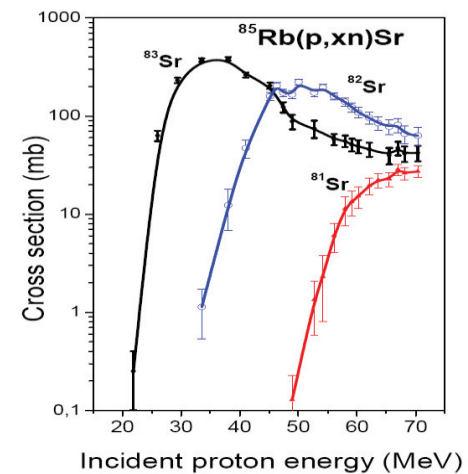
Status and basic parameters of electric power system displayed on the operator monitor



Status and basic parameters of the proton beam transport system displayed on the operator monitor



Targets for radiochemical production

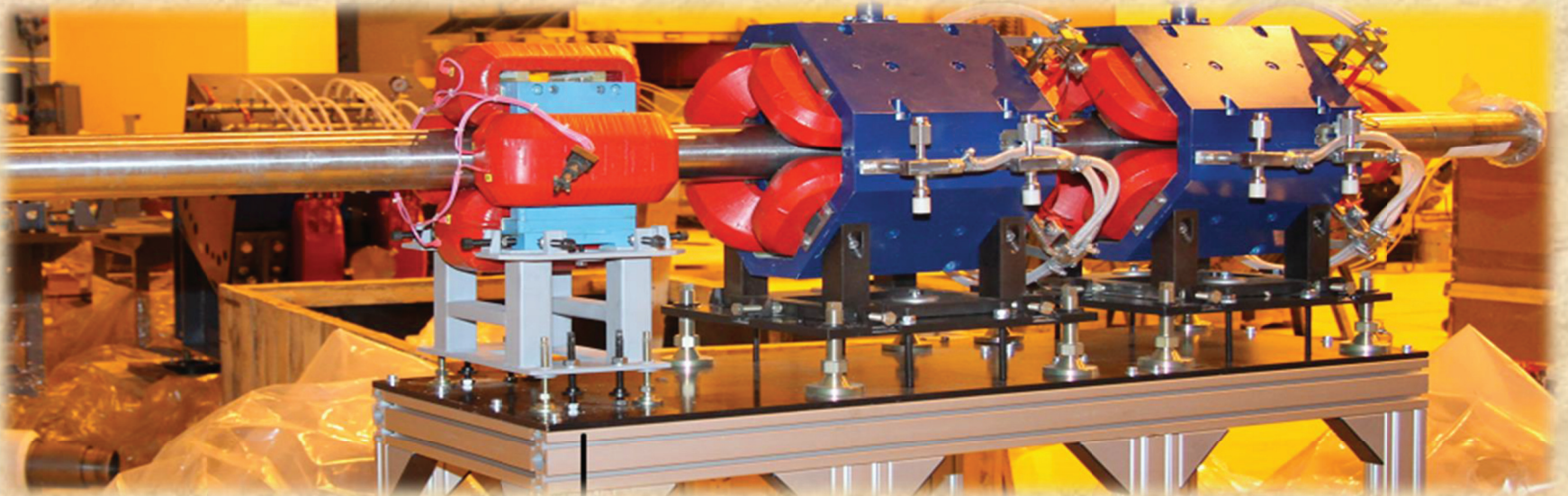
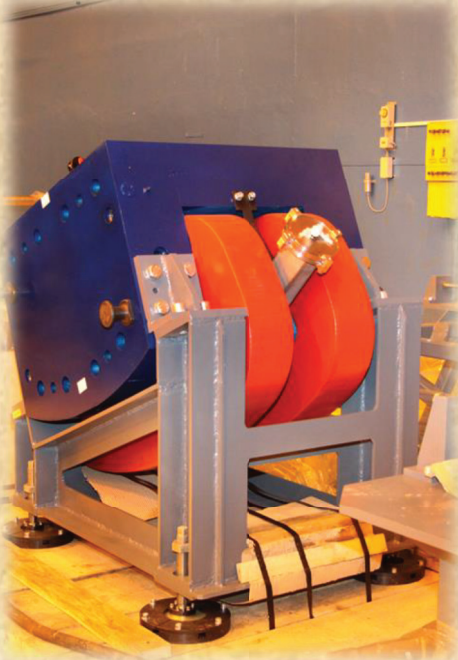


Sr isotopes' yield as a function of energy

Mass separator

Project of the proton beam transport system

Elements of the CC-80 beam transport system





Cyclotron and the first section of the beam transport system consisting of a matching, correcting, switching and bending magnets, doublet of quadrupole lenses and beam diagnostics are located in the experimental hall.

Conclusion:

Our short-term plans include attaining of the design intensity of 100 μA by using reserves untapped so far:

- completion of works on optimization of the electromagnetic field in the cyclotron central area;**
- increase of the injection current up to its specified value (from 900 up to 2500 μA)**
- use of the buncher potentialities with the anticipated effect of not less than two times higher current of the extracted beam; in works on the physical start-up, the buncher installed in the external injection system line was not used;**
- efforts made towards these lines guarantee the attaining of design parameters of the C-80 cyclotron system and will offer opportunities for further increase of the accelerated proton beam intensity.**



Thank you for your attention !